CPSC 250
Advanced Linear Regression Concepts

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1 Motivation

In earlier lectures, we considered simple linear regression involving one independent variable.

Real-world data analysis is often more complicated.

In this lecture, we discuss:

- multiple regression variables,
- parameter interpretation,
- correlations between variables,
- matrix representations,
- and data structures for regression.

2 Complex Regression Problems

Suppose we want to predict:

$$y$$

using several variables:

$$x_1, x_2, x_3, x_4$$

For example:

- y: housing price,
- x_1: square footage,
- x_2: zip code,
- x_3: number of bedrooms,
- x_4: number of bathrooms.

3 The Regression Equation

We seek an equation of the form:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

This is linear in the parameters.

4 Why This Is Difficult

Visualizing this problem directly would require a very high-dimensional plot.

For example:

- one dependent variable,
- four independent variables.

This is impossible to visualize completely.

Therefore, regression becomes both:

- a computational problem,
- and an interpretation problem.

5 Parameter Uncertainties

In practice, we determine:

$$\beta_0 \pm \Delta\beta_0$$

$$\beta_1 \pm \Delta\beta_1$$

and so on.

A parameter consistent with zero may indicate that the corresponding variable is not important.

For example:

> If the coefficient for number of bedrooms is consistent with zero, housing prices may not strongly depend on that variable.

6 Exploratory Data Analysis

Before fitting, we should first examine the data.

For example:

$$y \text{ vs. } x_1$$

$$y \text{ vs. } x_2$$

and so on.

This helps identify:

- trends,
- bad data,
- outliers,
- nonlinear behavior.

7 Correlations Between Variables

Independent variables may themselves be correlated.

For example:

$$x_2 \text{ vs. } x_3$$

might show a strong trend.

This can produce misleading results.

8 Correlation vs. Causation

An important warning:

> Correlation does not imply causation.

Two variables may move together without one causing the other.

Regression analysis must therefore be interpreted carefully.

9 Data Structures for Regression

There are two common choices:

1. NumPy arrays,
2. pandas DataFrames.

10 Matrix Interpretation

Suppose:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

for many data points.

We can write:

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{21}$$

$$y_2 = \beta_0 + \beta_1 x_{12} + \beta_2 x_{22}$$

and so on.

11 Matrix Form

These equations can be rewritten as:

$$\mathbf{y} = X\beta$$

where:

- $\mathbf{y}$ is the vector of dependent values,

- X is the design matrix,
- β is the parameter vector.

12 Linear Algebra Perspective

In principle:

$$\beta = (X^T X)^{-1} X^T \mathbf{y}$$

This is the mathematical foundation of many regression algorithms.

13 Important Practical Point

For most Python regression tools:

- X-data should be a two-dimensional matrix,
- y-data should be a one-dimensional array.

14 Simple NumPy Example

```
import numpy as np

xi = [1, 2, 3, 4, 5]

yi = [1.1, 2.1, 2.9, 3.9, 5.1]

X = np.array(xi).reshape((-1,1))

y = np.array(yi)
```

15 Why Reshape?

The line:

```
reshape((-1,1))
```

converts the array into a column matrix.
This is required because regression libraries expect:

$$N \times M$$

data structures for independent variables.

16 Printing NumPy Arrays

NumPy arrays display differently from Python lists.

For example:

```
print(y)
```

may display:

```
[1.1 2.1 2.9 3.9 5.1]
```

without commas.

17 Data Cleaning

Before fitting, we should:

- plot the data,
- look for missing points,
- identify outliers,
- think about appropriate models.

18 Key Takeaways

- Real regression problems often involve many variables.
- Regression coefficients have uncertainties.
- Correlations between variables can complicate interpretation.
- Regression is fundamentally a linear-algebra problem.
- Most regression libraries require two-dimensional X-data.
- Exploratory plotting is extremely important before fitting.